# Software Requirements Specification (SRS) and Comprehensive System Architecture Analysis for The ArchEd Web Platform

## 1. Executive Summary and System Overview

The digital transformation of specialized educational domains necessitates complex, highly integrated web platforms capable of supporting multifaceted pedagogical objectives. The system under analysis is the web platform for "The ArchEd," accessible via the primary domain thearachiever.com. This platform is an educational technology (Ed-Tech) infrastructure based in India, explicitly engineered to serve the discipline of Architecture and its adjacent academic communities.[1] The ontological framework of this platform significantly transcends traditional learning management systems (LMS). It incorporates advanced digital twin concepts, competitive assessment modules, comprehensive virtual environments, and cross-platform mobile synchronization.[3]

This exhaustive Software Requirements Specification (SRS) and system analysis delineates the architectural topology, page-by-page structural composition, interactive elements, and underlying functional requirements of the platform in its entirety. By synthesizing the visible front-end interfaces with the inferred back-end data management strategies, this document provides a holistic blueprint of the digital ecosystem. The primary operational directive of The ArchEd is to facilitate practical, technology-driven education that empowers architecture students to bridge academic knowledge gaps and realize their potential across personal, spatial, and global dimensions.[1] The platform operates not merely as a repository of knowledge but as a dynamic engine for creative actualization, demanding a highly resilient and scalable technological foundation managed by its technical partner, Thanneeru Software Technologies (TST).[3]

The subsequent sections of this report will meticulously deconstruct the platform, beginning with the foundational pedagogical ontology that drives its technical requirements, proceeding to a macro-level navigational analysis, and ultimately delivering an exhaustive, page-by-page functional specification. This analysis directly addresses the explicit requirement to catalog the total number of distinct web pages and detail their individual contents, structural layouts, and interactive behaviors.

## 2. Platform Purpose, Vision, and Pedagogical Ontology

To accurately specify the software requirements of an Ed-Tech platform, one must first comprehend the educational philosophies driving its user experience. The ArchEd is built upon

a highly specific vision and mission that dictates its required technical capabilities. The overarching strategic vision is to empower students through creativity and professional excellence, fostering an environment characterized by sustainability, diversity, and innovation.[2] This is achieved through the delivery of virtual learning, intensive workshops, and research-based design practices that emphasize leadership, teamwork, critical inquiry, and adaptability.[2]

The platform's instructional design is heavily influenced by prominent architectural theorists. The integration of their philosophies directly into the user experience mandates a system architecture that is visually sophisticated, highly interactive, and context-aware. The system must support diverse media types and immersive experiences to reflect these guiding principles [2]:

- **Le Corbusier:** The emphasis on the mastery of forms in light requires the platform to support high-fidelity visual rendering, likely leveraging WebGL or similar technologies for spatial representation.
- **Norman Foster:** The advocacy for future-oriented design informed by historical awareness necessitates a comprehensive digital library and archival system within the platform, allowing students to seamlessly reference historical blueprints while drafting modern solutions.
- **Nari Gandhi:** The championing of simplicity as the ultimate sophistication serves as a primary user interface (UI) and user experience (UX) requirement. The platform must avoid cluttered layouts, favoring minimalist, intuitive navigational paradigms.
- **B.V. Doshi:** Viewing architecture as a state of being requires the platform to foster an immersive environment, potentially realized through its Metaverse/Archiverse modules, where students do not just view designs but inhabit them digitally.
- **Raj Rewal:** Promoting contextual design that absorbs the essence of a location necessitates geospatial integration within the platform, allowing students to map their digital creations onto real-world coordinates and environmental constraints.

Furthermore, the mission of The ArchEd includes fostering critical thinking, promoting ethical global approaches, and directly addressing macroeconomic factors such as global warming and climate change.[2] This implies that the content management system (CMS) must be highly adaptable, supporting the ingestion and distribution of contemporary research papers, environmental impact simulators, and multicultural design collaborative tools. The platform follows a defined organizational workflow—Client Needs, Direction, Design, Develop, and Value—which must be directly mirrored in the user journey maps and project management modules available to the students.[2]

# 3. System Actor Profiles and Access Control Mechanisms

A robust software architecture requires the explicit definition of system actors—the entities

that interact with the system—and the boundaries of their permissions. While the exact backend schema is abstracted, the functional requirements derived from the platform's services necessitate at least four distinct user profiles, each governed by specialized access control lists (ACLs) and state management protocols.

| Actor Designation | Role Description and System Capabilities | Required Access and Interactions |
|---|---|---|
| **Unauthenticated Guest** | Exploratory users arriving via search engines or social media. They can view the homepage, read the pedagogical philosophies, and browse the overarching service catalog. | Read-only access to /index.html, /about.html, /what-we-do.html, and public policy pages. Can initiate the contact form. |
| **Registered Student (Understudy)** | The primary user demographic. These actors have created authenticated sessions (likely via JWT or similar token-based protocols). | Write access to competition submission portals (/compitition/register.php), read/write access to the Social Interface, and read/interaction access to Quizzes and Courses.[4] |
| **Jury / Evaluator** | Domain experts tasked with reviewing competition submissions. This role requires specialized dashboard interfaces to view high-resolution PDF portfolios and input scoring metrics.[9] | Read access to isolated, anonymized student submissions. Write access to scoring databases. Scheduled access linked to "Jury Panel Reveal" dates.[9] |
| **System Administrator** | Personnel from The ArchEd or Thanneeru Software Technologies (TST) managing the platform infrastructure.[3] | Full CRUD (Create, Read, Update, Delete) access to the CMS, user databases, payment gateway logs, and policy updates. |

The transition between these roles requires secure authentication gateways, particularly given

the financial transactions associated with competition registrations and the intellectual property inherently attached to architectural design submissions.

# 4. Global Topological Framework and Navigational Architecture

A structural analysis of thearachiever.com reveals a consistent global template applied across the domain, ensuring navigational continuity and a unified brand presence. The global UI taxonomy is segmented into three primary persistent elements that frame the user experience regardless of the specific URI being accessed.

## 4.1 Global Header and Primary Navigation

The header serves as the primary orienting mechanism and is injected into the Document Object Model (DOM) of every primary page. It is anchored by the organizational branding, prominently featuring "The ArchEd" nomenclature and associated graphical logo.[3] The primary navigation menu provides immediate, top-level access to the core informational hierarchies of the site, establishing the primary user funnels:

- **Home:** Routing to the root index.
- **About:** Routing to the pedagogical and historical overview.
- **What We Do:** Routing to the comprehensive service catalog.
- **Contact:** Routing to communication and support protocols.

## 4.2 The "All Services" Lateral Navigational Sidebar

A recurring architectural motif on internal functional pages (such as the Quizzes and Archiverse pages) is the presence of an "All Services" sidebar list.[3] This localized navigation functions as a highly efficient lateral movement mechanism. It allows users to pivot directly between core functional modules without the cognitive friction of returning to the primary header index or utilizing the browser's back button. This structural recurrence suggests a hub-and-spoke navigational model, where the 'What We Do' directory acts as the central hub, and the individual service pages represent the interconnected spokes. The sidebar uniformly includes hyperlinked routing to Competition, Quizzes, Archiverse, Courses, Guest lectures, and the Social Interface.[3]

## 4.3 Global Footer and Legal Repositories

The global footer acts as the terminal navigational aid and the primary legal repository for the platform. It ensures persistent access to contact mechanisms and regulatory compliance documentation across all user sessions, which is critical for establishing institutional trust and adhering to digital commerce regulations.[3] The data structures consistently rendered within the footer include the corporate address (Tirupati, AP-517501, India), direct communication links (Phone: +91 93728 25114, Email: info@thearachiever.com), and the copyright declaration (© 2023 The ArchEd. All right reserved.).[1] Crucially, the footer provides the technical attribution to Thanneeru Software Technologies (Powered By TST) and houses the routing to mandatory legal

documents: Terms & Conditions, Privacy policy, and Returns & Refund policy.[3]

# 5. Exhaustive Page Inventory and Identification

Addressing the core query regarding the scope and scale of the platform, a comprehensive crawl and analysis of the provided data yields a specific inventory of distinct web pages (URIs) that constitute The ArchEd's web presence. The platform operates on a multi-page application (MPA) or hybrid rendering model, with distinct endpoints mapping to specific functional domains.

The exhaustive list of the fourteen (14) identified pages comprising the platform's architecture is as follows:

| **Page Functionality** | **Identified URI Path** | **Primary Purpose** |
|---|---|---|
| **Homepage** | / or /index.html | Primary user acquisition, core messaging, and initial CTAs. |
| **Institutional Overview** | /about.html | Articulation of mission, vision, workflows, and philosophical foundations. |
| **Service Catalog** | /what-we-do.html | Central hub detailing the six core offerings of the platform. |
| **Contact Interface** | /contact.html | User support, dynamic forms, and corporate geographic details. |
| **Competition Hub** | /competition.html | Overarching information regarding architectural contests. |

| | | |
|---|---|---|
| **Registration Portal** | /compitition/register.php | The transactional and secure file-upload gateway for competitors. |
| **Assessment Module** | /quizzes.html | Interactive, competitive learning and retention testing. |
| **Virtual Environment** | /metaverse.html | The "Archiverse" framework for digital spatial creation. |
| **Distance Learning** | /courses.html | The digital instruction and video delivery interface. |
| **Expert Insights** | /guest-lecture.html | Short-form, high-impact video or live webinar content. |
| **Community Hub** | /social-interface.html | The architectural peer-to-peer networking platform. |
| **Usage Contract** | /terms.html | Legal stipulations governing user interactions and intellectual property. |
| **Data Governance** | /privacy_policy.html | Disclosures on telemetry, cookie tracking, and data security. |
| **Financial Policy** | /refund_policy.html | The economic remediation logic for courses and |

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|---|---|---|
| | | competitions. |

The following sections will dissect each of these fourteen pages, detailing their specific content, structural layouts, inferred backend logic, and interactive elements.

# 6. Page-by-Page Structural and Functional Specifications

## 6.1 The Homepage (/index.html)

The root domain represents the apex of the primary user acquisition funnel. It is structurally optimized for immediate impact, brand positioning, and the rapid segmentation of incoming traffic into actionable pathways.[1]

The layout is anchored by a high-impact, potentially rotating hero banner system or a multi-sectional landing sequence.[1] This module relies on high-resolution architectural imagery coupled with repetitive, semantically clustered marketing copy designed to elicit user engagement. The specific text strings displayed include variations of "Unleash Your Imagination with Architectural Excellence," "Designing Spaces that Inspire and Captivate," and "Inspiring Architecture for the Modern World".[1] Each of these visionary statements is paired with a distinct "Know More" Call-to-Action (CTA) button, which functionally routes the user deeper into the /about.html or /what-we-do.html directories.

A distinct anomaly within the hero messaging is the phrase "Answering Nature's call is unavoidable," which is paired with a specific "Sign UP" CTA.[1] Given the context of the platform, this is not a general registration prompt, but rather a targeted thematic funnel directing users toward the "PUBLIC TOILETS FOR THE PUBLIC COMPETITION 2024".[9] This indicates a dynamic homepage layout capable of prioritizing time-sensitive events and routing users directly into transactional workflows (/compitition/register.php).

Below the hero sections, the homepage features an explicit value proposition block detailing the organizational summary: "About The ArchEd. We are an ed-tech platform in India for Architecture. We strive to help every Architecture student to fill the gaps and realize their potential...".[1] This segment establishes the platform's focus on "protentional and practical education using technology," immediately signaling to the user that the site is an interactive tool rather than a static brochure. The page naturally concludes with the global footer, providing immediate access to the corporate address in Tirupati, the +91 phone number, and the info@ email address.[1]

## 6.2 The About Us Page (/about.html)

The 'About Us' interface functions as the ideological, methodological, and historical manifesto

of the organization.[2] The requirement for this page is not merely to display text, but to structure complex philosophical concepts into digestible, visually appealing modules. It is structurally divided into distinct semantic segments.

The first module is the **Institutional Overview**, which reiterates the platform's identity as a novel Ed-Tech framework. It expands upon the mission to provide technical expertise utilizing a unique philosophical maxim: "We put real people into virtual spaces and then put out digital things in real spaces".[2] This statement is a critical functional requirement indicator; it heavily foreshadows the platform's investment in the 'Archiverse' (/metaverse.html) and augmented/virtual reality integrations, demanding a UI that visually connects traditional architecture with digital rendering.

The next segment houses the **Vision and Mission Modules**. The Vision component dictates the requirement to showcase the four pillars of their educational approach: Competitions, Virtual learning, Workshops, and Research-based design practices.[2] The UI likely utilizes an icon grid or interactive accordion to display these points cleanly. The Mission component outlines the pedagogical responsibilities: fostering critical thinking, promoting ethical approaches, and addressing global warming.[2]

A highly specific and unique structural feature of the About page is the detailed articulation of the **Our Work Process** layout.[2] This is presented as a serialized, numbered progression that translates the abstract art of architecture into a standardized user journey:

1. **01 Client Needs:** Emphasizing the foundational phase of understanding constraints.
2. **02 Direction:** The vectorization of design parameters.
3. **03 Design:** The creative process of developing aesthetics, form, and function.
4. **04 Develop:** The procedural and technical drafting phase.[7]
5. **05 Value:** The terminal assessment phase measuring community and environmental benefit.

Finally, the page incorporates a **Philosophical Context** module, displaying inspirational quotes from architectural luminaries (Le Corbusier, Norman Foster, Nari Gandhi, B.V. Doshi, Raj Rewal).[2] The system likely renders these as highly stylized blockquotes or interactive carousels, linking the platform's modern technology back to classical architectural heritage.

## 6.3 The Service Catalog (/what-we-do.html)

This directory page acts as the central routing hub and informational dashboard for the platform's six core service pillars.[4] The structural requirement for this page involves a highly organized grid or list-based layout (e.g., a CSS Grid or Flexbox container) where each module is accompanied by an icon or thumbnail, a descriptive taxonomy, and a hyperlink to its dedicated sub-page.

The content requirements for this page dictate clear, concise explanations of each service:

- **Competition:** Described as events where contenders outperform rivals in various rounds,

demonstrating abilities, information, and imagination.[4]

- **Quizzes:** Positioned as a unique assessment method to provide feedback on knowledge and understanding.[4]
- **Archiverse:** Outlined as the fundamental systems required to support the creation of virtual worlds.[4]
- **Courses:** Defined as "distance discovery," allowing remote study and focus on design.[4]
- **Guest Lectures:** Focused on the "snappiness" of visiting speakers delivering concise, enlightening presentations.[4]
- **Social Interface:** Described as a digital platform explicitly designed as an architectural social stage for peer interaction.[4]

This page must be heavily optimized for internal linking (SEO), ensuring that users can easily transition from browsing the general catalog to engaging with specific platform tools.

## 6.4 The Competition Hub and Registration Portal (/competition.html & /compitition/register.php)

The competition module represents the most complex transactional subsystem within The ArchEd platform. It requires advanced form validation, secure payment processing, heavy file handling, and intricate temporal logic governed by the server chronometer.

While /competition.html likely serves as the marketing landing page displaying current and past contests, /compitition/register.php is the active gateway where data ingestion occurs.[8] The use of the .php extension indicates a server-side scripting environment handling the business logic before writing to a relational database (e.g., MySQL or PostgreSQL).

The primary data available pertains to the "PUBLIC TOILETS FOR THE PUBLIC COMPETITION 2024".[9] The system architecture for this page must flawlessly execute the following temporal phasing and gateway logic:

| Registration Phase | Active Date Range | Required System Behavior |
|---|---|---|
| **Early Registration** | 16th Sept 2024 - 15th Oct 2024 | Application of discounted early-bird pricing tiers in the payment gateway. |
| **Standard Registration** | 16th Oct 2024 - 15th Nov 2024 | Automated backend script transition to standard pricing parameters at 00:01 on Oct |

| | | |
|---|---|---|
| | | 16th. |
| **Late Registration** | 16th Nov 2024 - 15th Dec 2024 | Final fee escalation applied. |
| **Submission Deadline** | 22nd Dec 2024 | Hard deactivation of the file upload forms on /register.php; rejection of subsequent POST requests. |
| **Jury Panel Reveal** | 27th Oct 2024 | Automated CMS publication of jury member profiles and credentials on the front-end. |
| **Result Declaration** | 20th Jan 2025 | Database query triggering the front-end rendering of the winners' gallery. |

The backend financial logic must account for substantial disbursement tracking, as the platform offers significant monetary incentives: 1st Prize (Rs. 1 Lakh), 2nd Prize (Rs. 50k), 3rd Prize (Rs. 30k), and Top 10 Honorable Entries (Rs. 2k each).[9] The system must also automate the generation of digital certifications and manage the publication pipelines for winning entries.

Crucially, the data ingestion constraints on the registration form are exceptionally strict. The UI must provide a multi-part form capable of accepting complex architectural portfolios. The requirements dictate that entries must be presented on A2-sized drawings in landscape format, requiring three copies (Plans, Elevations, Sections) and a consolidated softcopy in PDF format.[12] This dictates a robust cloud storage integration (such as AWS S3 or Google Cloud Storage) capable of handling massive, high-resolution architectural schematics without causing browser timeouts. The frontend requires JavaScript validation to check file MIME types, calculate file sizes, and potentially analyze PDF dimensions prior to allowing the server upload, ensuring the database is not corrupted by malformed submissions.

## 6.5 The Assessment Interface (/quizzes.html)

The Quizzes page is characterized by its pedagogical mandate to provide unique, interactive assessment tools that offer valuable feedback on a student's knowledge.[10] Structurally, the layout utilizes the global header, a main content area demarcated by an <h2> heading ("Quizzes"), an illustrative graphical asset, and the standardized "All Services" lateral navigation sidebar.[10]

The architectural requirements for this module are driven by its defining characteristics: it must be "Interactive, Competitive, Versatile".[10]

- **Interactive:** The requirement for participants to "actively engage with the material and respond to questions" implies that standard HTML radio buttons are insufficient. The UI must support dynamic Document Object Model (DOM) manipulation, potentially including drag-and-drop architectural element matching, multi-select logic, and visually rich scenario-based questions.
- **Competitive:** This aspect suggests the presence of gamification elements. The system architecture must support real-time leaderboards, requiring low-latency database reads and potentially WebSocket integrations to update scores live as students complete modules.
- **Versatile:** The text explicitly defines the delivery vectors as "online test base, and mobile applications".[10] This is a profound architectural implication. It dictates a "headless" or decoupled backend configuration where the core quiz logic (questions, valid answers, scoring weights, user progress) is managed via a central RESTful or GraphQL API. This API is then independently consumed by the web front-end (React, Vue, etc.) and the respective iOS/Android native applications, ensuring state synchronization across all devices.

## 6.6 The Archiverse / Metaverse Environment (/metaverse.html)

The Archiverse page represents the documentation for the most technologically ambitious and resource-intensive subsystem of The ArchEd. It details the "fundamental system and framework that upholds the creation and activity of a virtual World or Universe".[3] The page follows a standard layout (Header, an <h2> "Archiverse" heading, descriptive paragraphs, a Metaverse visual asset, and the Service Sidebar) [3], but the technology it describes is highly complex.

The specifications outline an "intricate and diverse" design comprising several cooperating technological pillars [3]:

1. **Virtual World Stages:** These are the actual rendering environments. To provide the required "consistent and vivid experience" [3] directly within a web browser, the platform must utilize advanced WebGL or WebXR frameworks (such as Three.js, Babylon.js, or PlayCanvas). This allows the browser's GPU to render complex 3D architectural models without requiring external software installations.
2. **Man-made Consciousness (Artificial Intelligence):** The explicit integration of AI into a

spatial design platform suggests advanced features such as procedural generation of architectural forms, automated environmental stress testing (e.g., simulating lighting or structural loads), or intelligent non-player character (NPC) interactions within the digital space.[3] This requires immense server-side computational power and API bridges to external machine learning inference engines.

3. **Interoperability Protocols:** This is a crucial architectural insight. The documentation explicitly states these protocols are used to "empower the transfer of information and resources between various virtual Universes and stages in the Metaverse".[3] This means The ArchEd is not building a walled garden. The system must support standardized 3D asset formats (such as glTF, USDZ, or OBJ) and potentially decentralized ledger technologies (blockchain) or universal APIs to allow students to securely port their architectural models out of the platform and into broader, external global metaverse environments.

The documentation notes that the design of the Metaverse is "continually advancing as new advancements and stages arise," and that architects and engineers "keep on pushing the limits".[3] This dictates a highly modular, containerized codebase (e.g., using Docker and Kubernetes) capable of rapid, iterative deployment of new rendering techniques or protocol updates without disrupting the stability of the legacy platform.

## 6.7 Distance Learning and Courses Interface (/courses.html)

The Courses page serves as the gateway to the platform's synchronous and asynchronous digital instruction modules. Defined under the 'What We Do' umbrella as "distance discovery," this module allows architecture students to study and focus on design from remote locations over the internet.[4]

The systemic requirements for delivering high-quality architectural education online are stringent. The page must render a dynamic, searchable, and filterable course catalog. Behind the scenes, the system must support robust video streaming capabilities. Because architectural lectures often feature highly detailed CAD screen recordings or high-definition physical model tours, the platform must utilize a globally distributed Content Delivery Network (CDN) to prevent buffering latency and ensure adaptive bitrate streaming based on the student's bandwidth. Furthermore, the system requires Document Object sharing for supplementary materials and complex progress tracking mechanisms (such as SCORM or xAPI compliance) to precisely monitor student engagement, track module completion, and automatically trigger backend logic for generating and issuing digital certificates upon successful course conclusion.

## 6.8 Expert Insights and Guest Lectures (/guest-lecture.html)

This module is strategically optimized for high-impact, short-form educational content. The platform's emphasis on the "snappiness" of visiting speakers—referring to their ability to deliver concise yet enlightening presentations in a short period—indicates a reliance on carefully

edited video-on-demand (VOD) assets or highly structured live webinar integrations.[4]

The UI for this page requires a visually appealing gallery of speakers, complete with professional biographies, topical categorization tags, and intuitive scheduling interfaces for upcoming live sessions. The backend must be capable of integrating with live streaming APIs (such as Zoom, WebRTC, or custom RTMP servers) and managing calendar invitations and notification dispatches to registered students.

## 6.9 The Community Hub and Social Interface (/social-interface.html)

The Social Interface page explicitly provisions a digital platform specifically designed as an "architectural social stage for design students to interact".[4] This module fundamentally transforms The ArchEd from a one-directional broadcasting tool (where the platform simply serves content to the user) into a bidirectional, user-generated community hub.

The functional requirements for an architectural social interface are distinct from generic social networks:

- **Rich User Profiles:** Students require detailed digital portfolios to showcase their project history, competition results [9], and exported Archiverse assets.
- **Algorithmic Feed and Timeline Logic:** The system must curate peer activities, heavily optimized for rendering visual, high-fidelity architectural images and 3D embeds rather than simple text posts.
- **Interaction Primitives:** Mechanisms must be in place to allow users to like, comment, provide structured architectural critiques, and form collaborative study groups.
- **Database Implications:** Maintaining a real-time social network generates massive amounts of relational data. The architecture mandates the use of high read/write throughput databases (such as NoSQL solutions like MongoDB, or graph databases like Neo4j) to manage the rapid generation of activity logs, notifications, and complex relational connections between thousands of users.

## 6.10 The Contact and Support Infrastructure (/contact.html)

The Contact interface provides the primary vector for user support, administrative troubleshooting, and corporate inquiries.[11]

The static structural components of the page list the comprehensive corporate identifiers required for institutional legitimacy. This includes the physical address (20-2-472/9F, Revenue Ward No. 20, Korlagunta Main Road, Tirupati, Andhra Pradesh - 517501, India), telephonic support (+91 93728 25114), and email routing (info@thearachiever.com).[11] These elements are likely formatted using microdata schemas (like schema.org) to enhance local search engine visibility.

Dynamically, the platform features an active "Contact Form" component.[11] Standard architectural requirements for such a form include input fields for User Name, Email Address, a Subject taxonomy dropdown (to route queries to the correct internal department), and a

Message payload box. The presence of an asynchronous success state prompt ("Your message was sent successfully") [11] indicates that the form does not rely on outdated, full-page POST reloads. Instead, it utilizes AJAX (Asynchronous JavaScript and XML) or Fetch API protocols to submit the payload seamlessly to a backend SMTP relay or Customer Relationship Management (CRM) webhook, instantly verifying the submission and enhancing the user experience.

## 6.11 The Legal, Privacy, and Governance Frameworks

A comprehensive Ed-Tech platform that processes financial transactions (for courses and high-stakes competitions) and harvests personal telemetry data requires robust legal documentation. These documents are structurally mandated to exist on distinct URIs to ensure compliance with global digital commerce and privacy regulations.[10]

### 6.11.1 The Privacy Policy (/privacy_policy.html)

The Privacy Policy page reveals the explicit data capture methodologies and governance mechanisms engineered into the platform's architecture.[13]

- **Automated Telemetry Capture:** The platform's frontend automatically executes scripts upon connection to capture "Personal and Device Information." The system extracts the web browser user-agent strings, IP addresses (utilized for geographic localization and language setting), time zone configurations, and specific cookie identifiers installed on the device.[13]
- **Behavioral Analytics Routing:** As users traverse the site, the system continuously logs "Interaction Data." It tracks the specific web pages or architectural products viewed, the search terms or external websites that referred the user, and granular data on how the user interacts with DOM elements.[13] This necessitates the integration of a powerful backend analytics engine or the implementation of third-party tracking pixels (e.g., Google Analytics, Meta Pixel).
- **Data Utilization and Sovereignty:** The captured telemetry is routed for operational provision, system trend analysis, customer support communication, and the retargeting of advertising for platform services.[13] Crucially, the policy asserts a strict zero-sharing rule regarding Personal Information with external third parties.[13]
- **Compliance and Security:** The platform acknowledges that while "reasonable precautions" are taken, "absolute security cannot be guaranteed" over internet transmissions.[13] This serves as a standard legal liability shield. Furthermore, the provision of user rights to access, correct, update, or delete personal data [13] acts as compliance with frameworks like GDPR or India's DPDP Act. This legally mandates the existence of backend administrative tools capable of executing permanent database sanitization and record deletion requests.

### 6.11.2 Terms & Conditions (/terms.html) and Refund Policy (/refund_policy.html)

These nodes are functionally required to establish the binding legal contract between the

student and the platform. The Terms page (/terms.html) must outline acceptable use policies, community guidelines for the Social Interface, and critically, the intellectual property rights regarding the architectural designs submitted into the Archiverse or competitions.[3] The Refund Policy (/refund_policy.html) must hardcode the financial remediation logic governing the platform, detailing exact timelines and conditions under which competition entry fees or course tuition can be reversed, directly informing the logic programmed into the backend payment gateway.

# 7. Cross-Platform Integration and Mobile Architecture Synergy

The ArchEd ecosystem is not constrained solely to web browser interfaces operating via HTTP. A critical and highly complex component of its system architecture is its cross-platform mobile strategy. The platform extends its reach via dedicated, native applications available on the iOS App Store and the Google Play Store.[5] This multi-device approach drastically alters the required backend architecture, shifting it from a monolithic web server to a decoupled, API-driven microservices environment.

## 7.1 Apple iOS Environment Architecture

The iOS application operates under the specific namespace "ArchEd" and is accessible globally via the iTunes/App Store listing.[5] The deployment of this application requires strict adherence to Apple's Human Interface Guidelines and stringent review processes regarding data privacy and in-app purchasing protocols. The necessity for users to authenticate via an Apple ID ("If you are not logged-in the iOS appstore app, you'll be prompted for your your Apple ID and/or password") [5] before installation establishes a secure, trusted execution environment at the device level. The iOS front-end, likely coded in Swift or Objective-C, must maintain constant, secure synchronization via RESTful or GraphQL APIs with the core Thanneeru Software Technologies (TST) backend. This ensures that a student's course progress, quiz leaderboard rankings, and Social Interface interactions remain perfectly congruent whether they are accessing the platform via a desktop browser or an iPhone.

## 7.2 Android Operating System Architecture

Concurrently, the platform maintains an Android application distributed via the Google Play ecosystem.[6] The installation process involves explicit, granular permission granting ("A pop-up window with the permissions required by ArchEd will be shown. Click on Accept to continue the process").[6] Given the platform's heavy reliance on spatial mapping, digital reality, and the ambitions of the Archiverse, these requested permissions likely include extensive access to the mobile device's hardware cluster. This would encompass camera access for augmented reality overlays, local storage read/write permissions for downloading bulky, offline course assets or rendering architectural models, and continuous network state monitoring. The application, likely built using Kotlin or a cross-platform framework like React Native or Flutter, must handle device fragmentation across thousands of different Android screen sizes and hardware

capabilities, ensuring consistent pedagogical delivery.

### 7.3 The API Gateway and State Synchronization Imperative

The existence of three distinct graphical user interfaces (the Web MPA, the native iOS app, and the native Android app) communicating with a singular educational objective mandates a highly robust API Gateway architecture. When a user completes a "Competitive Quiz" [10] on the Android application while commuting, the resulting score, updated user profile, and system telemetry must be instantaneously ingested by the API, processed by the server, and reflected in the Web-based leaderboards without latency. This architecture ensures that the core business logic (authentication, user profile state, content delivery, transaction processing) is completely decoupled from the presentation layer, allowing rapid iteration of the UI without risking database corruption.

## 8. Non-Functional System Requirements and Infrastructure Demands

Beyond the structural layouts and explicit features of the fourteen identified pages, the underlying system must adhere to severe non-functional constraints to maintain operational viability, ensure data security, and deliver an acceptable user experience under load.

### 8.1 Compute Performance and Dynamic Scalability

The integration of a functional Metaverse/Archiverse [3] and the simultaneous handling of complex architectural design file uploads [9] demand massive server bandwidth and elastic compute power. The backend infrastructure, managed by TST [3], cannot rely on static shared hosting. It must utilize auto-scaling cloud infrastructure (such as AWS EC2 auto-scaling groups or containerized Kubernetes clusters) to dynamically allocate resources based on traffic spikes. This is particularly critical during the terminal phases of competition deadlines (e.g., the hours leading up to midnight on December 22nd, 2024, for the Public Toilets competition) [9], where hundreds of users may simultaneously attempt to initiate massive POST requests.

### 8.2 Geometric Storage Constraints and Optimization

The data ingestion requirements for platform competitions are immense. The mandate to ingest "Three copies of each A2-sized drawing... plus a softcopy in PDF format" [12] for every single competitor creates an exponential storage escalation curve. Standard relational databases will fail under this load. The system architecture must instantly transition heavy binary large objects (BLOBs)—such as the A2 PDFs and 3D Archiverse models—to specialized, cheap cloud object storage arrays (e.g., AWS S3 buckets). The system must only retain the relational metadata (the URI link to the file, the user ID, the timestamp) within the primary SQL/NoSQL databases to maintain rapid query efficiency and fast page load times across the platform.

### 8.3 Cryptographic Security and Transactional Integrity

The active collection of payment data for high-stakes competition registration (ranging from Rs. 2k to Rs. 1 Lakh in prize disbursements, and corresponding inbound registration fees) [9] necessitates strict PCI-DSS compliant payment gateway integrations (such as Stripe, Razorpay, or similar regional providers). The platform cannot store raw credit card data on its own servers. Furthermore, the explicit admission of active telemetry capture, behavioral tracking, and device fingerprinting [13] mandates end-to-end Transport Layer Security (TLS) encryption across the entire domain, effectively enforcing strict HTTPS protocols on all URI routing to prevent man-in-the-middle data interception.

# 9. Conclusion: Architectural Synthesis and Future System Trajectory

The exhaustive structural analysis and requirement specification of The ArchEd platform reveals a highly sophisticated, multi-layered digital ecosystem designed to fundamentally disrupt traditional architectural pedagogy. The system is an amalgamation of e-commerce transactional engines, real-time social networking, and advanced spatial computing.

The platform executes a remarkable synthesis of classical theoretical architecture and modern computer science. By explicitly mapping the design philosophies of visionaries like Norman Foster and B.V. Doshi [2] onto digital structures like the Archiverse [3], the system transcends a mere repository of courses. It manifests as a functional digital laboratory where architecture students practice "research and practice-based design" [2] within AI-augmented virtual stages.

The structural reliance on Thanneeru Software Technologies (TST) [3] provides a necessary enterprise-grade backend. This foundation enables the platform to operate highly complex, time-gated financial and submission systems capable of handling massive file throughput.[9] The deliberate fragmentation of content delivery across Web, iOS, and Android interfaces [5], coupled with a centralized, data-rich telemetry capture system [13], points toward an ecosystem built for ubiquitous, continuous computing.

The software requirements extracted from the platform's current state project a trajectory focused heavily on interoperable spatial web technologies. As the "interoperability conventions" [3] of the Metaverse mature, the required system architecture will undoubtedly evolve to support real-time, multi-user architectural rendering directly within the browser ecosystem. The ArchEd is not merely delivering education; it is actively architecting the intricate digital infrastructure required to train the next generation of spatial and environmental designers.

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